

**COMSATS UNIVERSITY ISLAMABAD,
LAHORE CAMPUS**



**Object Oriented Programming
Lab Task**

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Submitted to:

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Date:

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LAB TASK 1

```
C:\Users\Administrator>javac Student.java Task1Demo.java

C:\Users\Administrator>java Task1Demo
Alishba Javed SP26-BAI-001 12
Minahil Sehar SP26-BAI-002 15
Abeer Amina SP26-BAI-003 18
After changing s2 the outputs are:
Alishba Javed SP26-BAI-001 12
Minahil Sehar SP26-BAI-002 20
Abeer Amina SP26-BAI-003 18

C:\Users\Administrator>
```

Task1Demo - Notepad

File Edit Format View Help

```
class Task1Demo{
    public static void main (String[] args){
        Student s1 =new Student();
        Student s2 =new Student();
        Student s3 =new Student();

        s1.studentId ="SP26-BAI-001";
        s1.name = "Alishba Javed";
        s1.completedCredits=12;

        s2.studentId ="SP26-BAI-002";
        s2.name ="Minahil Sehar";
        s2.completedCredits=15;

        s3.studentId ="SP26-BAI-003";
        s3.name = "Abeer Amina";
        s3.completedCredits=18;

        System.out.println(s1.name +" " +s1.studentId+" " +s1.completedCredits);
        System.out.println(s2.name +" " +s2.studentId+" " +s2.completedCredits);
        System.out.println(s3.name +" " +s3.studentId+" " +s3.completedCredits);

        s2.completedCredits=20;

        System.out.println("After changing s2 the outputs are:");

        System.out.println(s1.name +" " +s1.studentId+" " +s1.completedCredits);
        System.out.println(s2.name +" " +s2.studentId+" " +s2.completedCredits);
        System.out.println(s3.name +" " +s3.studentId+" " +s3.completedCredits);
    }
}
```

LAB TASK 2

Administrator: Command Prompt

```
C:\Users\Administrator>javac Student.java Task2Demo.java

C:\Users\Administrator>java Task2Demo
Before adding credits:
ID:SP26-BAI-001, Name: Abeer Amina, Credits: 30
ID:SP26-BAI-002, Name: Alishba Javed, Credits: 25
After adding credits:
ID:SP26-BAI-001, Name: Abeer Amina, Credits: 35
ID:SP26-BAI-002, Name: Alishba Javed, Credits: 35

Remaining Credits:
Abeer Amina :95
Alishba Javed :95

C:\Users\Administrator>
```

Student - Notepad

File Edit Format View Help

```
class Student{
    String studentId;
    String name;
    int completedCredits;

    void addCredits(int credits){
        completedCredits= completedCredits +credits;
    }
    int remainingCredits(int degreeCredits){
        int remaining =degreeCredits -completedCredits;
        return remaining;
    }

    String summary(){
        return "ID:" + studentId + ", Name: "+ name +", Credits: "+completedCredits;
    }
}
```

Task2Demo - Notepad

File Edit Format View Help

```
public class Task2Demo{
    public static void main(String [] args){
        Student s1= new Student();
        Student s2= new Student();
        Student s3= new Student();

        s1.studentId="SP26-BAI-001";
        s1.name ="Abeer Amina";
        s1.completedCredits =30;

        s2.studentId="SP26-BAI-002";
        s2.name ="Alishba Javed";
        s2.completedCredits =25;

        System.out.println("Before adding credits:");

        System.out.println(s1.summary());
        System.out.println(s2.summary());

        s1.addCredits(5);
        s2.addCredits(10);

        System.out.println("After adding credits:");
        System.out.println(s1.summary());
        System.out.println(s2.summary());

        int remaining1= s1.remainingCredits(130);
        int remaining2= s2.remainingCredits(130);

        System.out.println("\n Remaining Credits: ");
        System.out.println(s1.name + " :" +remaining1);
        System.out.println(s2.name + " :" +remaining2);
    }
}
```

LAB TASK 3

```
C:\Users\Administrator>notepad OverloadDemo.java
```

```
C:\Users\Administrator>javac Overload.java
```

```
error: file not found: Overload.java
```

```
Usage: javac <options> <source files>
```

```
use --help for a list of possible options
```

```
C:\Users\Administrator>javac OverloadDemo.java
```

```
C:\Users\Administrator>java OverloadDemo
```

```
Enrolled in course: CSC241
```

```
Enrolled in CSC241, Section: 2
```

```
Enrolled in numeric course: 241
```

OverloadDemo - Notepad

File Edit Format View Help

```
public class OverloadDemo {
    void enroll(String courseCode) {
        System.out.println("Enrolled in course: " + courseCode);
    }

    void enroll(String courseCode, int section) {
        System.out.println("Enrolled in " + courseCode + ", Section: " + section);
    }
    void enroll(int numericCourseCode) {
        System.out.println("Enrolled in numeric course: " + numericCourseCode);
    }
    public static void main(String[] args) {
        OverloadDemo demo = new OverloadDemo();
        demo.enroll("CSC241");
        demo.enroll("CSC241", 2);
        demo.enroll(241);
    }
}
```

LAB TASK 4

```
C:\Users\Administrator>javac Student.java PassByValue.java
```

```
C:\Users\Administrator>java PassByValue
```

```
EXPERIMENT A
```

```
Before: 10
```

```
Inside changeNumber before: 10
```

```
Inside changeNumber after: 99
```

```
After: 10
```

```
EXPERIMENT B
```

```
Before: 30
```

```
Inside changeStudent before: 30
```

```
Inside changeStudent after: 99
```

```
After: 99
```

```
EXPERIMENT C
```

```
Before: Original - 40
```

```
Inside replaceStudent: Temporary - 50
```

```
After: Original - 40
```

```
C:\Users\Administrator>_
```

 *PassByValue - Notepad

File Edit Format View Help

```
public class PassByValue {
    static void changeNumber(int x) {
        System.out.println("Inside changeNumber before: " + x);
        x = 99;
        System.out.println("Inside changeNumber after: " + x);
    }
    static void changeStudent(Student st) {
        System.out.println("Inside changeStudent before: " + st.completedCredits);
        st.completedCredits = 99;
        System.out.println("Inside changeStudent after: " + st.completedCredits);
    }
    static void replaceStudent(Student st) {
        st = new Student();
        st.name = "Temporary";
        st.completedCredits = 50;
        System.out.println("Inside replaceStudent: " + st.name + " - " + st.completedCredits);
    }
}

public static void main(String[] args) {
    int number = 10;
    System.out.println("EXPERIMENT A: ");
    System.out.println("Before: " + number);
    changeNumber(number);
    System.out.println("After: " + number);

    Student student = new Student();
    student.name = "Abeer";
    student.completedCredits = 30;

    System.out.println("\nEXPERIMENT B");
    System.out.println("Before: " + student.completedCredits);
    changeStudent(student);
    System.out.println("After: " + student.completedCredits);

    Student original = new Student();
    original.name = "Original";
    original.completedCredits = 40;

    System.out.println("\nEXPERIMENT C");
    System.out.println("Before: " + original.name + " - " + original.completedCredits);
    replaceStudent(original);

    System.out.println("After: " + original.name + " - " + original.completedCredits);
}
}
```